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Computer Methods in Combustion

**Project Report: Thermodynamic
Simulation of an Internal
Combustion Engine with Wiebe
Combustion Model and Cantera
Chemical Kinetics**

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1 Introduction

The development of modern internal combustion (IC) engines requires a deep understanding of both macro-scale thermodynamic processes and micro-scale chemical phenomena. Numerical simulations play a crucial role in reducing development costs and optimizing engine efficiency while minimizing toxic emissions.

The primary objective of this project is to develop a comprehensive zero-dimensional (0D) thermodynamic model of a reciprocating IC engine. The model integrates the classic Wiebe mass fraction burned function with precise advanced chemical kinetics utilizing the Cantera software library. By simulating the CH_4 /air combustion process under spark-ignition (SI) conditions, the model bridges the gap between empirical cylinder pressure analysis and theoretical chemical equilibrium states.

2 Literature Review

The thermodynamic analysis of internal combustion engines typically relies on the ideal gas law and the application of the First Law of Thermodynamics to a closed system. Traditionally, empirical functions are used to approximate the rate of heat release. The most widely accepted approach was introduced by I.I. Wiebe, who formulated an exponential relation describing the mass fraction of burned fuel as a function of the crank angle.

While the Wiebe model provides an excellent macroscopic description of heat release, it lacks chemical fidelity. In reality, parameters such as the adiabatic flame temperature (T_{ad}) and laminar flame speed (S_L) govern the boundary limits of engine operation, thermal stresses, and emission profiles. Utilizing detailed kinetic mechanisms, such as the GRI-3.0 mechanism containing 53 species and 325 elementary reactions, allows for the accurate prediction of equilibrium product composition and combustion speed. This integrated approach combines classical engineering thermodynamics with computational chemistry.

3 Model Description

3.1 Piston-Crank Kinematics

The instantaneous cylinder volume $V(\varphi)$ as a function of the crank angle φ is derived from the full exact kinematics of the slider-crank mechanism:

$$V(\varphi) = V_c + V_d \cdot \frac{(1 - \cos \varphi) + \frac{1}{\lambda_r} \left(1 - \sqrt{1 - \lambda_r^2 \sin^2 \varphi}\right)}{2} \quad (1)$$

where V_c is the clearance (combustion chamber) volume, V_d is the displacement volume, and $\lambda_r = r/l$ represents the crank radius to connecting rod length ratio.

3.2 Wiebe Combustion Function

The mass fraction of burned fuel $x(\varphi)$ represents the progress of chemical energy release inside the cylinder:

$$x(\varphi) = 1 - \exp \left(-a \cdot \left(\frac{\varphi - \varphi_0}{\Delta \varphi} \right)^{m+1} \right) \quad (2)$$

where:

- $a = 6.908$ is the efficiency parameter corresponding to 99.9% fuel burnout,
- φ_0 is the start of combustion (crank angle),
- $\Delta\varphi$ is the total combustion duration,
- m is the Wiebe form factor (shape parameter reflecting the engine type).

3.3 Thermodynamic Framework (First Law)

The pressure change within the cylinder closed system is governed by the differential form of the First Law of Thermodynamics:

$$\frac{dp}{d\varphi} = \frac{\gamma - 1}{V} \frac{dQ}{d\varphi} - \frac{\gamma p}{V} \frac{dV}{d\varphi} \quad (3)$$

where γ is the ratio of specific heats (C_p/C_v), and dQ represents the incremental chemical heat input, computed as:

$$dQ = \frac{dx}{d\varphi} \cdot Q_{total}, \quad Q_{total} = m_{fuel} \cdot H_u \cdot \eta_i \quad (4)$$

Here, H_u represents the lower heating value of the fuel, and η_i accounts for the indicated efficiency (thermal losses).

3.4 Chemical Kinetics via Cantera

Cantera is utilized to solve the multi-component chemical equilibrium. The fresh mixture conditions (T_1 , p_1) and the equivalence ratio $\Phi = 1/\lambda$ are passed to the GRI-3.0 mechanism. The equilibrium state is calculated using an isobaric-isoenthalpic solver (HP equilibrium), returning T_{ad} and the molar fractions of the exhaust gas constituents. Additionally, a one-dimensional free flame model (**FreeFlame**) is solved to determine the laminar burning velocity S_L .

4 Results and Discussion

4.1 Baseline Simulation Indicators

The numerical simulation was executed for a baseline Spark-Ignition (SI) configuration ($\varepsilon = 10.0$, $\lambda = 1.0$, $\varphi_0 = -10.0^\circ$ CA). The integrated thermodynamic indicators and chemistry limits are summarized in Table 1.

Table 1: Thermodynamic and Chemical Indicators

Parameter	Value	Unit
Heat Input (Q)	1692.69	J
Indicated Work (W)	889.00	J
Maximum Pressure (p_{max})	60.59	bar
Maximum In-cylinder Temperature (T_{max})	3252.2	K
Adiabatic Flame Temperature (T_{ad})	2235.3	K
Laminar Burning Velocity (S_L)	42.18	cm/s
Indicated Mean Effective Pressure (IMEP)	17.780	bar
Thermal Indicated Efficiency (η_{th})	52.52	%

4.2 Exhaust Gas Equilibrium Composition

The detailed multi-species equilibrium analysis conducted in Cantera for the stoichiometric mixture provided the molar composition of the exhaust gases. Species with a molar fraction higher than 0.1% are presented in Table 2.

Table 2: Equilibrium Product Composition ($\lambda = 1.0$)

Chemical Species	Molar Fraction [-]
N_2	0.7083
H_2O	0.1832
CO_2	0.0850
CO	0.0094
O_2	0.0048
H_2	0.0037
OH	0.0030
NO	0.0020

The presence of dissociation products such as CO , H_2 , and OH , along with a distinct concentration of nitric oxide (NO), indicates the explicit inclusion of high-temperature chemical equilibrium in the model.

4.3 Graphical Diagnostics

The simulation generates a multi-panel diagnostic chart presented in Figure 1.

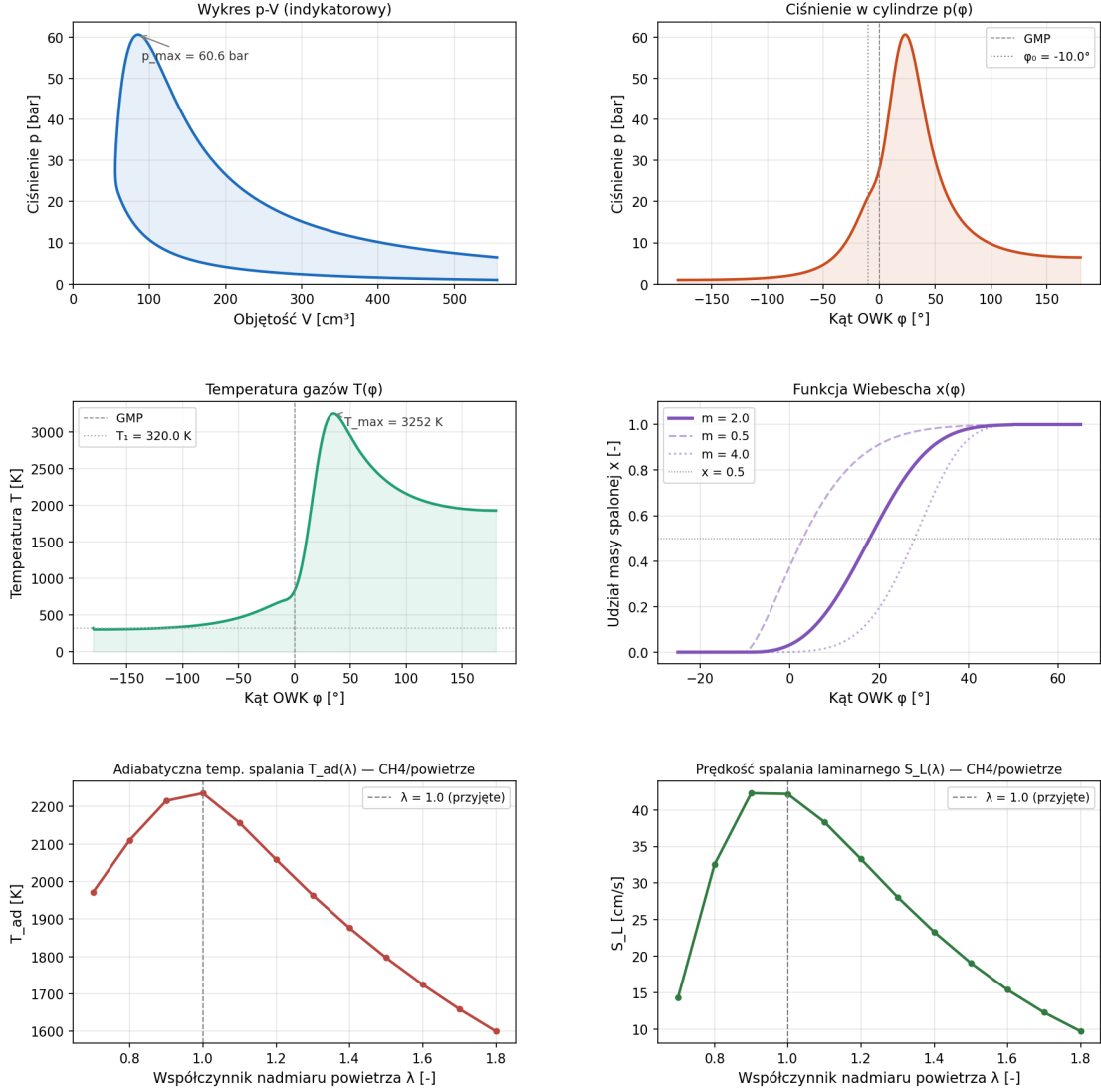


Figure 1: Multi-panel thermodynamic and chemical diagnostic charts for the simulated engine.

The indicator p-V diagram loop directly corresponds to the total work output. A major point of discussion is the comparison between temperatures: the peak in-cylinder temperature T_{max} (3252.2 K) significantly exceeds the Cantera T_{ad} value (2235.3 K). This is physically consistent because T_{ad} is computed using the uncompressed inlet conditions (T_1, p_1) , whereas the actual combustion occurs in a highly preheated, compressed mixture near the Top Dead Center (TDC), combining the effects of compression work and chemical heat release.

5 Conclusions

The developed 0D numerical model successfully coupled phenomenological engine simulation with detailed multi-species chemical kinetics. The following conclusions can be

drawn:

1. The Wiebe function effectively describes macroscopic heat release, allowing robust pressure computation via the finite difference method.
2. Cantera integration provides a realistic boundary constraint for temperature and flame kinetics, proving that full fuel-air chemistry depends strictly on the excess air ratio λ .
3. The model accurately predicts high-temperature dissociation products (CO , NO), which are vital for subsequent emissions optimization.

References

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